Moonshot Blueprint: what a new architecture must satisfy

ARC White-Box Estimation Challenge 2026, Phase 2 · foyshal_sk · last updated 24 Sep 2026, evening (Bangladesh time, UTC+6) · submissions close 18 Oct 05:59 Bangladesh time (17 Oct 23:59 UTC).
This is the moonshot's design specification. It records what our experiments have established, which paths are closed, and which conditions a new architecture must meet. Detailed evidence is in the Complete Notes PDF, part K3 (section numbers in brackets). The idea list is in the Experiment Queue PDF.

1. Target and distance

Score = final-layer MSE × max(0.1, FLOPs / 2^{41}). Unit: 1u = 2^{31} FLOPs, so util 0.1 ≈ 102u.

Rank (24 Sep)	Team	Adjusted	Raw MSE	Util	Cost (u)
1	J2W	2.1e-9	1.64e-8	0.131	~134
2	suliman_tadros	2.2e-9	1.63e-8	0.138	~141
3	marius_binner	2.3e-9	1.56e-8	0.147	~151
4	a_s6	2.5e-9	2.00e-8	0.126	~129
33	**us (#332110)**	**5.1e-9**	**2.11e-8**	**0.244**	**~249**

Raw MSE needed for a top-3 score (adjusted ≈ 2.25e-9) at each cost:

Cost (u)	250 (now)	205	154	133	123	110	≤102 (floor)
Raw MSE needed	9.2e-9	1.12e-8	1.50e-8	1.73e-8	1.87e-8	2.10e-8	2.25e-8

Key point: accuracy alone cannot reach the top 3: it would need raw 9e-9, and no one is near that. **At our current accuracy, cutting the cost to ~110u (44% of today's) reaches the top 3, and ≤102u ties #1.** a_s6 shows near-our accuracy (2.0e-8) at ~129u. The moonshot target is therefore **the same accuracy at ≤ ~115u**. Every further u saved buys accuracy slack (table above).

2. How far we are: an honest assessment

Stage	Status
1. Error map (where accuracy is lost)	✅ Complete. Every channel's contribution is measured (section 4).
2. Cost ledger (where FLOPs go)	✅ Complete. Every matmul is attributed to its source line (section 5).
3. Closed paths	✅ Largely done. The whole linear-compression family is closed by an argument, and temporal tricks are closed too (sections 6–7).
4. Find a new mechanism	⏳ **Not found yet.** Round 10 (young tier) and Round 11 (change what is represented) are out.
5. One-layer falsifier passes	⬜ Pending
6. Prototype on one MLP, score < 0.8 × chain	⬜ Pending
7. Kaggle 100-MLP paired check → submit	⬜ Pending

We are at **stage 4**. The search space is now small: only a few families remain (section 8), and ~23 days are left. The tooling for stages 5–7 is ready: dense captures, teacher arrays, the paired harness and the Kaggle runners. Once a mechanism exists, a falsifier takes minutes, a prototype 1–2 days, and a 100-MLP check ~5 hours.

3. The current method (what must be beaten)

The V29/V31 factorised K3 chain (504aldo, MIT) carries the mean, the covariance C and κ3 at every layer. κ3 is held as a **sum over sources**: every ReLU gives birth to a source with legs P and A = P·a_b, where a_b = diag(Φ)C_{off}.

- The young tier (ages ≤ 4) keeps dense legs; older sources share a basis (rank 384 / 224); κ4 is memoryless (λ law).
- Means and covariances come from the bivariate Wick/Edgeworth closure.

4. Error map: where accuracy is lost (K3ab–K3ad)

- **C1.** Both closures are exact: given the true inputs, the mean closure is off by 9e-11. All error comes from the closure's **inputs**.
- **C2.** Channel oracles (true value injected at every layer; reduction in final MSE): **variance −47%**, **κ4 diagonal −30%** (acts only through the mean closure), **κ3 diagonal −14%**, **all three −94%**.
- **C3.** The error comes from the freshly born **all-distinct** part of κ3 at every layer, i.e. the post-ReLU κ3(h_i,h_j,h_k).
  - That part is 9–16% wrong relative to itself, and it causes 91–96% of the D21 slice error.
  - The slice error grows from 4% to 7% with depth and is random (slope 1.00, R^2 0.996).
  - The κ4 multi-index inputs add further error.
- **C4.** Errors accumulate through the mean (gain 0.65–0.96 per layer) and through the covariance. The variance error is really the error of the off-diagonal post-ReLU covariance, with ~4.2e-5 of fresh error per entry per layer.

C12. The chain's own per-channel error (the stage-1 bar for any proposal), free-running plain chain vs the 6e7-sample teacher, MLP #0, relative Frobenius error (teacher sampling-noise floor in brackets); experiments/probes/r11_chain_channel_ref.py:

Layer	mean (rms, units of σ)	variance	d3 = $κ3_{iii}$	κ4 diagonal (slope, corr)	C off-diagonal	D21 off-diagonal
7	1.9e-4	0.04% (0.02%)	3.2% (1.1%)	15% (3.6%) (0.95, 0.78)	0.39% (0.17%)	4.6% (1.3%)
11	3.1e-4	0.09% (0.02%)	3.8% (0.6%)	22% (2.5%) (0.89, 0.78)	0.59% (0.13%)	5.7% (0.8%)
14	4.3e-4	0.12% (0.02%)	4.1% (0.4%)	32% (2.0%) (0.79, 0.81)	0.70% (0.11%)	6.2% (0.5%)
15	4.6e-4	0.14% (0.02%)	4.6% (0.4%)	32% (2.0%) (0.78, 0.83)	0.73% (0.10%)	6.6% (0.5%)
16	5.1e-4	0.14% (0.02%)	4.7% (0.4%)	34% (1.8%) (0.75, 0.82)	-	-

All deep-layer errors are well above the teacher's noise. **The bar is severe: variance to ~0.1% and off-diagonal covariance to ~0.7%.** This is why sampling fails and why κ3 must be carried at all. The weakest channel is the κ4 diagonal: 32–34% wrong at depth, with a systematic slope of 0.75–0.79 (the chain's κ4 is ~22% too small along the truth). Yet the κ4 constants test at their score optimum (C5), so the missing part is pattern, not scale.

C13. Diagonal-only closures are dead (Round 11 answers; r11_diag_closure_gate.py). One step from the TRUE post-ReLU state, the next layer's d3 is 90–99% wrong when propagated from the diagonal alone (Hadamard powers), 83–98% with a covariance-driven (dressed-tree) term, and still 22–29% with the exact full repeated-index slice + tree (chain free-run: 3.2–4.7%). **d3 is fed almost entirely by off-diagonal κ3**, so the diagonal channels cannot be carried on their own. Slice-state recursions (Copilot RSC: carry Y21 + a response matrix, fitted coefficients) are 36% wrong on D21 even in-sample from the true state. Also: best rank-1 D21 is 35–49% wrong; diag + rank-64 covariance is still 12–33% wrong off-diagonal (bar 0.7%).

C14. Structure of the κ4 diagonal (r11_verify.py V5). The next layer's κ4 diagonal is ~90% the (2,2) pair term $3\Sigma Q_p^2 Q_q^2 K22_{post}(p,q)$; the (4) and (3,1) terms are 1–8%. Even with the exact post-ReLU (4), (3,1), (2,2) cumulants, the next κ4 diagonal is 14 / 22 / 33 / 43 / 46% wrong (L4/8/12/15/16): the (2,1,1) + all-distinct (1,1,1,1) remainder grows with depth and is larger than the chain's own error (32–34%). The chain's (2,2) model wk4m is 15–28% wrong (pattern corr 0.65–0.73). **A κ4 fix needs the higher post-ReLU structure — again memory, not a per-neuron rule.**

5. Cost ledger: where the FLOPs go (K3ah)

Component	Cost (u)	Note
Young-source transport	~54	ages 1–4, dense n×n legs
Young-source D21 contraction	~50	products of the form $(X \circ P)Y^T$
Join (a source entering the shared basis)	~41–46	range finder / eigen
Old tier (shared basis)	~41	rank 384 / 224
Thin feedback legs	~15	
Strassen additions	~12	
Covariance + closure	~8	cheap
Total	**~249**	88% matmul

C5. One-for-one law: every knob inside the chain (age gates, ranks, passes, screening, join rebuild, exact old tier, R_FB) trades ~1% of cost for ~1% of accuracy, in either direction. Examples:

- exact old tier: MSE −3.9%, cost +19%
- R_FB 128: MSE −3.1%, cost +43%
- join transport: cost −6.6%, MSE +1–6%

The chain sits at its own optimum. **Halving the cost is impossible inside this family.**

C6. Covariance and closure cost only ~8u. Almost all the cost is carrying κ3. A Gaussian (κ2-only) chain is very cheap but ~10× worse (dropping the old tier alone: ×35).

6. Structure of the κ3 memory (K3aj, measured 24 Sep)

Test	Result	Meaning
Horizon oracle: keep only sources of age ≤ h	L15 error: h2 84%, h4 65%, h6 46%, h8 31%, h10 **16%**	The memory is **long**: 10-layer-old sources still matter.
Age-Gram (similarity between ages)	mean cosine 0.29–0.39; first mode ~88% of the total; r = 6 modes still 8.5%	Coherent, but not low-rank across ages.
Shift-stable carrier (L14 → L15)	eps 2.9%, rho 0.944; carrier + birth rebuild the total D21 to only 13.9%	Not a pass. Uniform weights (the total κ3 itself) are transported **exactly** anyway.
Source-free dressed tree (DHMP)	36–40% error on the next D21 (gate 2.5%); tree vs. true all-distinct part: corr 0.43–0.54	About half the all-distinct part is **memory** and cannot be rebuilt from pair state.

C7. There is no temporal problem: the total κ3 moves as one object under one linear map, exactly. **The problem is purely spatial**: holding one dense n^3 tensor cheaply.

C8. The mathematical wall (GPT 5.6 Sol): no linear representation of the total κ3 can be both smaller and closed under transport by every dense M. Sym^3 is an irreducible GL(n) representation, so such a state needs ≥ $C(n+2,3) \approx 1.8\times10^8$ numbers. Tucker, TT, CP, vector-metric, age modes and shared bases all hit this wall.

- Our caveat: M is fixed and known, so the real bound is what future readouts can observe, ≤ n^2 per remaining layer. At mid depth this is close to the chain's ~$3kn^2$.
- The lazy cubic circuit (store only birth matrices and M, read D21 by reverse mode) costs ≈ 245u, the same as the chain. The two are dual contraction orders of one tensor network.

Conclusion: changing how the tensor is factorised gains nothing. What is represented must change. (Note: the "same cost" statement counts the two known exact orders; it is not a proven arithmetic lower bound.)

7. How old sources reach the final means (K3m)

- **C9.** **~86%** of the old-source slices' effect travels through **mean shifts** at the next 1–2 layers (variance → E[relu]). That path is near-range and full-rank; only ~14% goes through the low-rank final-variance path. Adjoint "keep only the relevant directions" is therefore dead.
- **C10.** Dropping either the old D21 (n^3) or the old d3 (diagonal, n^2) at one late layer exceeds the error budget. The diagonal's weight grows toward the output.
- **C11.** The Khatri–Rao wall: Hadamard products of factored legs have rank r(r+1)/2, which reaches n for r ≥ 45. In practice the spectrum is flat and noise-like, so sketching would need d ≈ 10^8.

8. Remaining paths (in priority order)

#	Path	Why it could work	Falsifier / gate	Status
M1	**Channel-direct**: compute the channels the next means need (variance, d3, κ4 diagonal, and the covariance entries that feed them) without carrying the κ3 tensor	C8 says carrying the tensor cannot be cheaper; the final score depends only on a few per-neuron channels (C2).	≤125u; on teacher data, one-layer channel error ≤ the chain's	Round 11 out. Astra: no mechanism (honest no); visible channels alone are provably insufficient in general (hidden all-distinct perturbation), so any M1 scheme needs extra state carrying the all-distinct memory at ≤~15 products per layer
M2	**Young tier at half cost** (~104u → ≤55u)	Algebra may be left: A = P·a_b, a shared propagator for consecutive sources, closed forms for ages 1–2	≤55u at ≤2% relative D21 error	Round 10 out. Exact contraction-order search (Sonnet 5): the shipped order is already at the floor (105 u reproduced; reverse/mixed orders +75–150%). Needs a new representation, not a new order
M3	**Nonlinear, trajectory-specific statistic** (specific to one MLP, not a linear representation)	The C8 wall covers only linear states closed under every M	Set once a concrete proposal exists	Open
M4	Accuracy side: fix the fresh all-distinct κ3 and the κ4 pattern	Oracles: κ4 −30%, variance −47%	MSE −10%+ at no extra cost (mind C5)	Partial (FB_SCALE 1.25 shipped)
M5	Small gains: join aug64 (Q16), FB × beta grid	score −0.5% on the teacher networks	100-MLP paired < 1 − 2 SE	Running on Kaggle

9. Checklist for any new proposal

1. Does it hit the C8 wall, i.e. is it just another factorisation of the same κ3 tensor? If so, reject it.
2. Where does the old-source content go (C9: sources 10+ layers old, ~88% coherent)? Dropping it costs 36–40% error (DHMP).
3. Count the cost in n^3 products: ≤ ~115u? (One dense 1024^2 product ≈ 0.5u with Strassen L5, so ~230 products = 115u.)
4. Is there a one-layer falsifier on stored data? Teacher: pre/post mean, C, Y21 and κ4 diagonal at layers 3/7/11/14/15/16. Captures: every source at L7/11/14/15.
5. Grader constraints: 8 GB RAM, float32, residual time < 0.4 s, only flopscope + whestbench + stdlib, no print.

10. Dead paths (do not retry)

Family	What was measured
Tucker / TT / CP / ISDF / vector-metric compression	2% needs rank ≈ 250–300, i.e. 500u+ per layer (C8 explains why)
Diagonal-only κ3/κ4 transport, low-rank covariance, rank-1 κ3, slice-state recursion RSC (Round 11)	d3 one-step 83–99%; RSC 36% in-sample (C13)
Short horizon (FHQR), age pooling, age modes	h = 10 still 16%; r = 6 still 8.5%
Shift-stable carrier	2.9%, and the total κ3 is already carried exactly
Source-free tree (DHMP)	36–40%
Deterministic particles (moment-matched designs), aggregate rank-k CP of old κ3 (Round 11)	particles: variance 3–7%, D21 74–160% at L15, no better than random; CP needs rank ≫ 250 (unfolding bound)
Sampling / control variate	10^4 samples cost 156u and give the variance only to 1.4% (0.1% needed); a control variate needs corr ≥ 0.997
Adjoint low-rank path	only 14% of the effect is low-rank
Learned per-neuron correction, copula, 13-feature regression, star sources, V39 depth multiplier, pairwise κ4 transport	no gain on 100 MLPs or on the teacher
Join rebuild, age gates, ranks, passes, screening	one-for-one law (C5)

11. Timeline (Bangladesh time)

Date	Goal
24–28 Sep	Triage Round 10/11 answers; run every proposal's falsifier on stored data
~2 Oct	One mechanism passes a one-layer falsifier (otherwise shift time to M4/M5)
~8 Oct	Prototype on one MLP, score < 0.8 × chain
~12 Oct	Kaggle 100-MLP paired check
15–17 Oct	Submit and nominate the final two submissions